

v1.0Strong Force Deep Dive Request

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: One-Quark Strong Force Architecture **Type:** Scientific Change Log | **Date:** June 2026
Applies to: v1.0Strong Force Deep Dive Request V2.0.md

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines’ Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^{+7} particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
alpha_s precision	0.1179 (V1.0)	PDG 2024: 0.1180±0.0009. 5-loop running now available. V2.0 runs alpha_s from any scale	LOW	Modest precision improvement and 5-loop vs 4-loop running
Glueball candidates	Not in V1.0	BESIII 2023: X(2370) pseudoscalar glueball. LQCD: 1710-1780 MeV lightest. GLUEBALL type added	HIGH	Confirmed new particle class absent from V1.0
QGP critical point	T_c only	HotQCD 2024: CEP at mu_B~440 MeV, T~107 MeV estimated. V2.0 adds finite-density phase diagram	MEDIUM	QCD phase diagram completion with CEP — new simulation regime
Confinement string tension	sigma ~ 0.18 GeV ²	BMW 2024: sigma = 0.1815±0.0012. Casimir scaling added: octet = 9/4*fundamental	LOW	Precision improvement + color-factor dependent string tension

Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** Glueball
 - **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
 - **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
 - **New interaction modes:** Triggered by new experimental discoveries
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Unchanged Architecture

All foundational philosophy is **carried forward unchanged:** - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric ($C \cdot F \cdot S$ product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0 Strong Force Deep Dive Request V1.0 archived | V2.0 is the active specification as of June 2026